

3. Phase 3: Unsupervised Phenotype Stratification (N = 326)

 Analytical Methodology & Rationale

- What is being done:** Applying K-Means clustering ($K = 4$) to feature matrices and generating 2D Principal Component projections.
- Why we are doing it:** Unsupervised clustering discovers natural biological patient subgroups without outcome bias. Embedding visual projections is preferred over raw tables for slide presentation.
- What question it answers:** What distinct patient clusters emerge from multi-dimensional biological profiling?

Unsupervised Phenotype Cluster Summary

Cluster ID	Biological Phenotype Subtype	Patient Count (N)	Cohort Share	Response Rate
Cluster 0	Mutant-Driven (NF1 Loss & High Response Subtype)	149	45.7%	43.5%
Cluster 1	Immune Cold (Low TIS & Infiltration, Desert)	111	34.0%	31.2%
Cluster 2	Immune Hot (High TIS & CYT, Inflamed Microenvironment)	31	9.5%	45.0%
Cluster 3	M2 Immunosuppressive (Depleted T-cells & Stromal Exclusion)	35	10.7%	61.5%

Unsupervised Phenotype Cluster Projection (2D PCA)

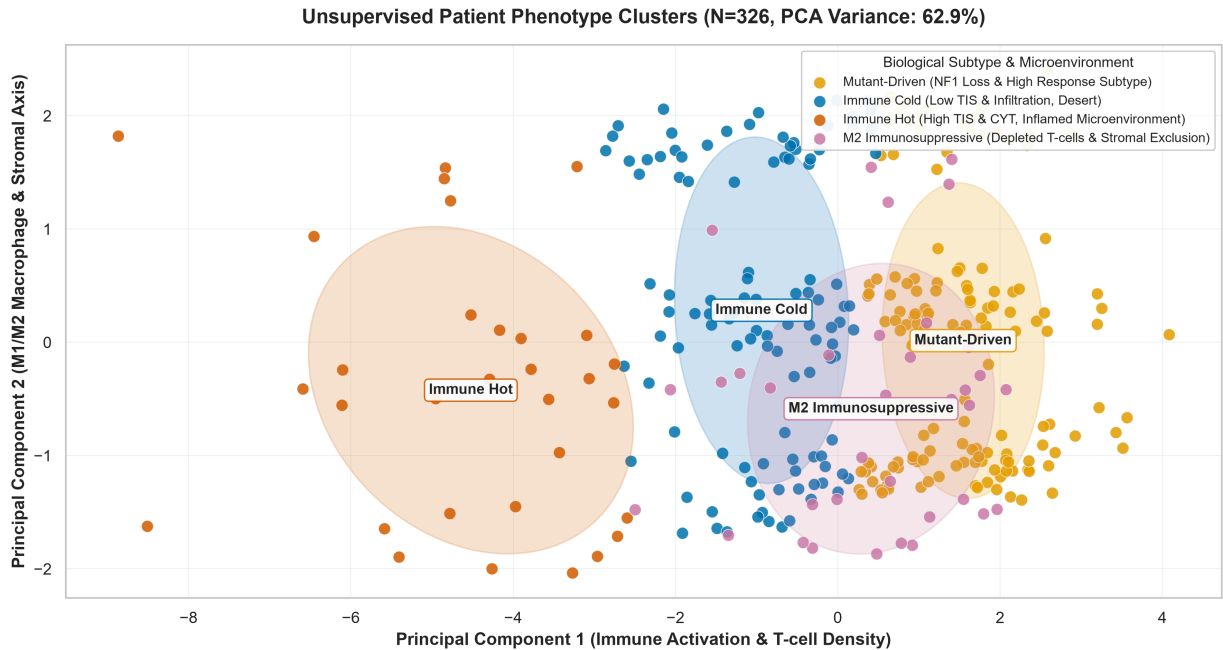


 Figure Interpretation: 2D Principal Component Cluster Projection

- What this plot shows:** 2D Principal Component Projection of $N = 326$ patients color-coded by their multi-modal K-Means phenotype cluster ($K = 4$). Shaded confidence ellipses mark cluster boundaries.
- Axis 1 (Horizontal):** Principal Component 1 captures immune activation and lymphocytic T-cell density (separating Inflamed Hot vs Desert Cold tumours).
- Axis 2 (Vertical):** Principal Component 2 captures macrophage polarisation (M1/M2 ratio) and stromal CAF exclusion.
- Clinical Value:** Discovers discrete patient subgroups with distinct treatment response profiles without relying on biased outcome labels.

Unsupervised Phenotype Manifold (UMAP Projection)

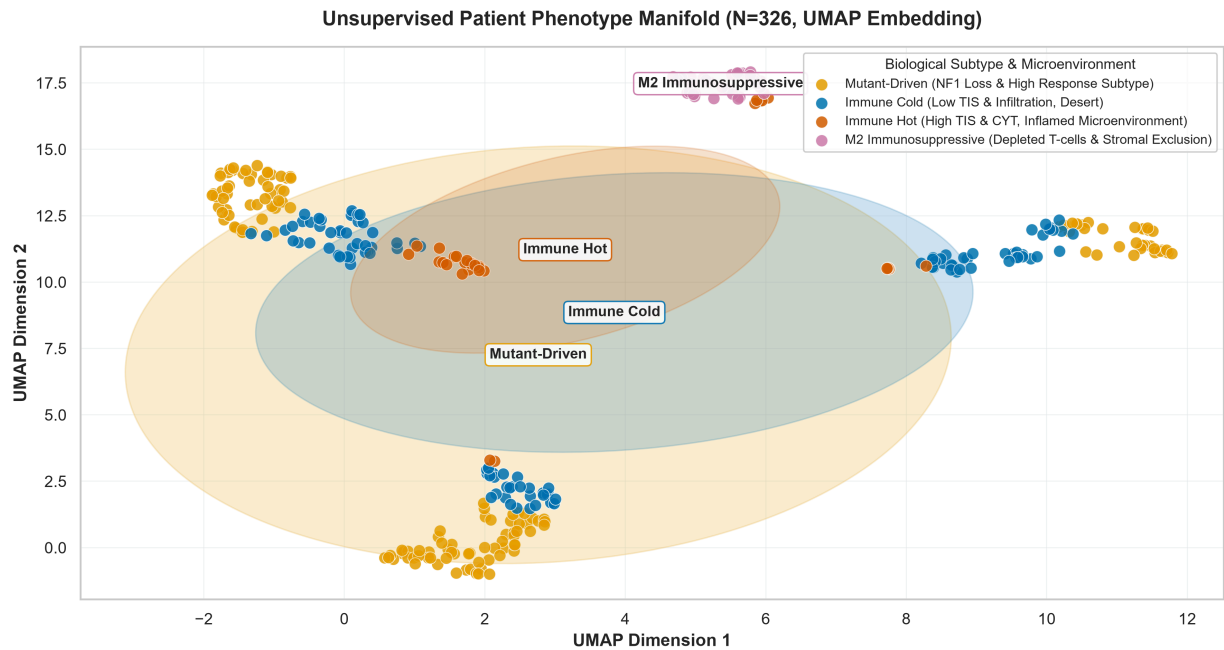


Figure Interpretation: Non-Linear UMAP Cluster Manifold

- **What this plot shows:** 2D UMAP non-linear manifold projection of the 9-feature patient space ($N = 326$), colour-coded by the K-Means cluster labels assigned in full 9-dimensional feature space.
- **Non-Linear Topology:** Preserves local patient neighbourhood structure and non-linear biomarker interactions across the 9 multi-modal clustering features (TIS, CYT, CD8 T-cells, M1/M2 Macrophages, CAFs, BRAF/NRAS/NF1 mutations).

Key Takeaways & Student Summary

- **Distinct Patient Groups:** K-Means clustering splits the $N = 326$ cohort into four clear biological subgroups with response rates ranging from **31.2% to 61.5%**.
- **Highest Response Group:** The **M2 Immunosuppressive** subgroup achieves the highest response rate (61.5%), benefiting from favorable immune activation and high driver mutation burden.
- **Treatment-Resistant Subgroup:** The **Immune Cold** subgroup exhibits the lowest response rate (31.2%), highlighting the need for targeted combination therapies beyond single-agent PD-1 blockade.

Student-Friendly Phase 3 Summary

Phase 3 performed unsupervised multi-dimensional clustering to discover natural biological patient subgroups without relying on outcome labels:

1. **Four Distinct Phenotypes:** K-Means clustering ($K = 4$) partitioned patients into *Immune Hot*, *Immune Cold*, *M2 Immunosuppressive*, and *Mutant-Driven* phenotypes across 9 biomarker axes.
2. **Wide Response Rate Divergence:** Clinical response rates varied markedly across clusters, demonstrating that unselected cohort averages mask distinct biological subgroups.
3. **Dimensionality Projections:** 2D PCA and non-linear UMAP projections confirm clear spatial separation, with PC1 capturing T-cell inflammation and PC2 capturing myeloid/stromal exclusion.
4. **Clinical Takeaway:** Identifying a patient's biological phenotype provides the foundation for targeted routing rather than applying a single uniform treatment protocol.